KAIZEN MATHS

Lesson starter

Partial Fractions: Decomposition

Repeated Linear Factors

Name _____

S1

Express in partial fractions:

$\frac{6x+11}{(x+2)^2}$

S2

Express in partial fractions:

$\frac{6x-38}{(x-5)^2}$

S3

Express in partial fractions:

$\frac{-2x-15}{(x+5)^2}$

S4

Express in partial fractions:

$\frac{-5x+8}{(x-3)^2}$

Show your working

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KAIZEN MATHS

Exit ticket

Partial Fractions: Decomposition

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Name _____

E1

Express in partial fractions:

$\frac{6x-15}{(x-2)^2}$

E2

Express in partial fractions:

$\frac{4x-16}{(x-3)^2}$

E3

Express in partial fractions:

$\frac{-3x-7}{(x+5)^2}$

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Teacher answers & working

Partial Fractions: Decomposition · Repeated Linear Factors

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S1

Express in partial fractions:

$\frac{6x+11}{(x+2)^2}$

Answer:

$\frac{6}{(x+2)} - \frac{1}{(x+2)^2}$

Working:

Step 1: For a repeated linear factor, include both powers of the factor.

$\frac{6x+11}{(x+2)^2} = \frac{A}{(x+2)} + \frac{B}{(x+2)^2}$

Step 2: Multiply through by $(x + 2)^2$ to remove the denominators.

$6x + 11 = A(x + 2) + B$

Step 3: Expand the right hand side.

$6x + 11 = Ax + 12 + B$

Step 4: Compare coefficients of x: A = 6.

Step 5: Compare constants: 11 = 12 + B, so B = -1.

Step 6: Substitute A and B back into the repeated-factor form.

Therefore $\frac{6x+11}{(x+2)^2} = \frac{6}{(x+2)} - \frac{1}{(x+2)^2}$.

S2

Express in partial fractions:

$\frac{6x-38}{(x-5)^2}$

Answer:

$\frac{6}{(x-5)} - \frac{8}{(x-5)^2}$

Working:

Step 1: For a repeated linear factor, include both powers of the factor.

$\frac{6x-38}{(x-5)^2} = \frac{A}{(x-5)} + \frac{B}{(x-5)^2}$

Step 2: Multiply through by $(x - 5)^2$ to remove the denominators.

$6x - 38 = A(x - 5) + B$

Step 3: Expand the right hand side.

$6x - 38 = Ax - 30 + B$

Step 4: Compare coefficients of x: A = 6.

Step 5: Compare constants: -38 = -30 + B, so B = -8.

Step 6: Substitute A and B back into the repeated-factor form.

Therefore $\frac{6x-38}{(x-5)^2} = \frac{6}{(x-5)} - \frac{8}{(x-5)^2}$.

Teacher answers & working

Partial Fractions: Decomposition · Repeated Linear Factors

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S3

Express in partial fractions:

$\frac{-2x-15}{(x+5)^2}$

Answer:

$\frac{-2}{(x+5)} - \frac{5}{(x+5)^2}$

Working:

Step 1: For a repeated linear factor, include both powers of the factor.

$\frac{-2x-15}{(x+5)^2} = \frac{A}{(x+5)} + \frac{B}{(x+5)^2}$

Step 2: Multiply through by $(x + 5)^2$ to remove the denominators.

$-2x - 15 = A(x + 5) + B$

Step 3: Expand the right hand side.

$-2x - 15 = Ax - 10 + B$

Step 4: Compare coefficients of x: A = -2.

Step 5: Compare constants: -15 = -10 + B, so B = -5.

Step 6: Substitute A and B back into the repeated-factor form.

Therefore $\frac{-2x-15}{(x+5)^2} = \frac{-2}{(x+5)} - \frac{5}{(x+5)^2}$.

S4

Express in partial fractions:

$\frac{-5x+8}{(x-3)^2}$

Answer:

$\frac{-5}{(x-3)} - \frac{7}{(x-3)^2}$

Working:

Step 1: For a repeated linear factor, include both powers of the factor.

$\frac{-5x+8}{(x-3)^2} = \frac{A}{(x-3)} + \frac{B}{(x-3)^2}$

Step 2: Multiply through by $(x - 3)^2$ to remove the denominators.

$-5x + 8 = A(x - 3) + B$

Step 3: Expand the right hand side.

$-5x + 8 = Ax + 15 + B$

Step 4: Compare coefficients of x: A = -5.

Step 5: Compare constants: 8 = 15 + B, so B = -7.

Step 6: Substitute A and B back into the repeated-factor form.

Therefore $\frac{-5x+8}{(x-3)^2} = \frac{-5}{(x-3)} - \frac{7}{(x-3)^2}$.

Teacher answers & working

Partial Fractions: Decomposition · Repeated Linear Factors

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E1

Express in partial fractions:

$\frac{6x-15}{(x-2)^2}$

Answer:

$\frac{6}{(x-2)} - \frac{3}{(x-2)^2}$

Working:

Step 1: For a repeated linear factor, include both powers of the factor.

$\frac{6x-15}{(x-2)^2} = \frac{A}{(x-2)} + \frac{B}{(x-2)^2}$

Step 2: Multiply through by $(x - 2)^2$ to remove the denominators.

$6x - 15 = A(x - 2) + B$

Step 3: Expand the right hand side.

$6x - 15 = Ax - 12 + B$

Step 4: Compare coefficients of x: A = 6.

Step 5: Compare constants: -15 = -12 + B, so B = -3.

Step 6: Substitute A and B back into the repeated-factor form.

Therefore $\frac{6x-15}{(x-2)^2} = \frac{6}{(x-2)} - \frac{3}{(x-2)^2}$.

E2

Express in partial fractions:

$\frac{4x-16}{(x-3)^2}$

Answer:

$\frac{4}{(x-3)} - \frac{4}{(x-3)^2}$

Working:

Step 1: For a repeated linear factor, include both powers of the factor.

$\frac{4x-16}{(x-3)^2} = \frac{A}{(x-3)} + \frac{B}{(x-3)^2}$

Step 2: Multiply through by $(x - 3)^2$ to remove the denominators.

$4x - 16 = A(x - 3) + B$

Step 3: Expand the right hand side.

$4x - 16 = Ax - 12 + B$

Step 4: Compare coefficients of x: A = 4.

Step 5: Compare constants: -16 = -12 + B, so B = -4.

Step 6: Substitute A and B back into the repeated-factor form.

Therefore $\frac{4x-16}{(x-3)^2} = \frac{4}{(x-3)} - \frac{4}{(x-3)^2}$.

Teacher answers & working

Partial Fractions: Decomposition · Repeated Linear Factors

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E3

Express in partial fractions:

$\frac{-3x-7}{(x+5)^2}$

Answer:

$\frac{-3}{(x+5)} + \frac{8}{(x+5)^2}$

Working:

Step 1: For a repeated linear factor, include both powers of the factor.

$\frac{-3x-7}{(x+5)^2} = \frac{A}{(x+5)} + \frac{B}{(x+5)^2}$

Step 2: Multiply through by $(x + 5)^2$ to remove the denominators.

$-3x - 7 = A(x + 5) + B$

Step 3: Expand the right hand side.

$-3x - 7 = Ax - 15 + B$

Step 4: Compare coefficients of x: A = -3.

Step 5: Compare constants: -7 = -15 + B, so B = 8.

Step 6: Substitute A and B back into the repeated-factor form.

Therefore $\frac{-3x-7}{(x+5)^2} = \frac{-3}{(x+5)} + \frac{8}{(x+5)^2}$.